ISCF-BSCA: Prefix-Consistent Unified Multi-Horizon Forecasting

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Abstract

Keywords:

1. Introduction

Multi-horizon forecasts support decisions over multiple planning ranges, from short-term control to long-term scheduling. Yet most long-term time-series forecasting models and benchmark protocols remain horizon-specific: a separate model is trained for each forecast horizon H, such as 96, 192, 336, and 720 steps [? ? ?]. Recent work, including ElasTST and time-series foundation models such as TimesFM and Time-MoE, has begun to support varied or flexible forecasting horizons [? ? ?]. Nevertheless, such efforts remain sparse relative to the extensive horizon-specific literature, and varied-horizon forecasting is still insufficiently developed as a unified problem with an explicit task definition, systematic problem analysis, and targeted decoder design. In this work, we formulate these requirements explicitly and investigate how a unified forecaster should organize output-side representations across different parts of the future domain.

The horizon-specific protocol fragments forecasting into independent systems. As illustrated in Fig. 1a, models trained for different horizons can assign different values to the same future time step, despite identical history and forecast origin. Their outputs therefore need not form coherent, nested views of one future trajectory. Serving multiple horizons also requires separate training, storage, deployment and maintenance.

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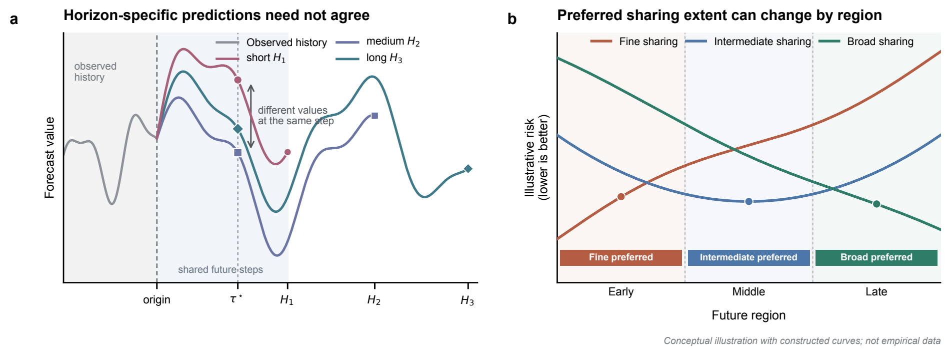

Figure 1: Two challenges in varied-horizon forecasting. **a**, Horizon-specific predictors may disagree at the same future time step $\tau^\star$ despite identical observed history. **b**, The sharing extent associated with the lowest risk can vary across future regions.

We therefore formulate varied-horizon forecasting as learning a single horizon-agnostic mapping from observed history and future-step index to prediction. Under this formulation, a future-step prediction depends on the history and its step index, but not on the requested horizon. For any $H_1 < H_2$, predictions over steps $1{:}H_1$ therefore remain identical under both requests. We call this basic requirement **cross-horizon prefix consistency (CHPC)**.

CHPC defines how forecasts at different horizons should relate, but not how a decoder should generate individual future regions. Most architectural advances focus on history encoding, while the output stage often uses one uniform mechanism to generate all future steps. Such a mechanism fixes how broadly a history-conditioned latent state is shared before step-specific prediction. Broad sharing can capture persistent trajectory structure, but may smooth local variations. Finer sharing offers greater step-specific flexibility, but provides weaker structural regularization. The preferred balance can vary across samples, variables and future regions, making one fixed sharing extent inadequate for the entire forecast domain. Fig. 1b summarizes this intuition, which we term **future-region sharing-demand heterogeneity**. We examine this problem in greater detail in Section 3.

Motivated by this heterogeneity, we propose ISCF-BSCA, an output-side decoder that integrates forecasts generated under different sharing extents. Independent Scope-Conditioned Forecasting (ISCF) represents each sharing extent through an independent history projection within a single scope-indexed forecast field. Each scope determines how broadly a history-conditioned latent state is reused before step-specific synthesis. A single-scope decoder applies one sharing extent throughout the forecast domain. ISCF instead integrates scope-conditioned forecasts through a target-conditioned allocation for each sample, variable and future step. The resulting forecast can adapt its sharing composition across future regions while retaining a single horizon-agnostic prediction function. To support joint learning, Balanced Scope Co-Adaptation (BSCA) supplies direct prediction signals to all scopes and discourages premature allocation concentration. BSCA operates only during training, adds no inference parameters or paths, and preserves CHPC across supported horizons.

Our contributions are threefold. First, we formulate varied-horizon forecasting as a unified system in which CHPC is a basic requirement, and identify future-region sharing-demand heterogeneity as an output-side challenge. Second, we introduce ISCF, which integrates forecasts generated under multiple sharing scopes through target-conditioned allocation. Third, we develop BSCA to support balanced scope learning without increasing inference-time complexity. Experiments across datasets from multiple application domains show that a single unified model outperforms separately trained horizon-specific forecasters. Component-wise ablations confirm the effectiveness of each component, while backbone transfer studies demonstrate decoder portability.

2. Related Work

2.1. Fixed-horizon multi-step forecasting

Multi-step forecasting has traditionally assumed a fixed forecast horizon and differs in how predictions within that horizon are generated. Recursive forecasting iterates one one-step model and feeds predictions back as inputs, which can accumulate errors [**?**]. Direct forecasting instead trains a separate model for each future step, avoiding recursive propagation but increasing estimation variance and weakening dependencies among future outputs [**?**]. The MIMO strategy predicts the complete future vector with one multi-output model, whereas DIRMO divides it into blocks generated by multiple models [**? ? ?**]. These strategies primarily trade error propagation, estimation variance, output dependence and computational cost within a prespecified horizon.

Modern long-term forecasting benchmarks largely adopt MIMO-like fixed-window prediction. DLinear, PatchTST and iTransformer map a history representation to one configured prediction length, with standard evaluations training separate models for horizons 96, 192, 336 and 720 [? ? ?]. This protocol optimizes each horizon independently but does not require agreement on overlapping future targets. We extend fixed-horizon strategy design to varied-horizon forecasting, where one model serves multiple request endpoints and returns prefix-consistent views of one predicted trajectory.

2.2. Unified and varied-horizon forecasting

Advances in time-series foundation models have encouraged models that operate across forecasting tasks or output lengths. TimesFM uses decoder-only pretraining across datasets, granularities and horizons, while Timer and Time-MoE generate flexible forecast lengths autoregressively [? ? ?]. UniTS further accommodates heterogeneous time-series tasks through task tokenization [?]. Although these models support multiple horizons or task specifications, they primarily pursue universal pretraining, task unification or scalable generation rather than a systematic treatment of varied-horizon forecasting. ElasTST is a notable exception. It combines structured attention masks, tunable RoPE, multi-scale history patches and horizon reweighting to improve robustness across inference horizons [?]. Its multi-scale mechanism, however, varies history-patch resolution rather than the output-side extent of state sharing. Our work targets this complementary problem by formalizing CHPC and modeling how state-sharing demand changes across future regions.

2.3. Forecast generation and output-side modeling

Most recent forecasting architectures concentrate on extracting and representing information from the observed history. Their output stages often remain shallow fixed-window projections. DLinear directly maps temporal inputs to future steps, while PatchTST and iTransformer project patch-level or variate-level representations through linear prediction heads [? ? ?]. Consequently, the forecasting phase is commonly treated as a readout of an increasingly sophisticated history Encoder rather than a separate modeling problem.

Beyond shallow output projections, several studies have introduced structured mechanisms for forecast generation. TFFS combines common and step-specific future features, N-HiTS synthesizes hierarchically interpolated components, and Implicit Forecaster composes predicted waves into a global trajectory [? ? ?]. These methods establish forecast generation as a first-class design problem rather than a passive readout. However, their cross-step sharing structure is predefined by the decoder topology. ISCF instead makes state-sharing extent explicit and constructs region-local forecasts under multiple scopes before step-specific synthesis.

2.4. Multi-scale forecasting and adaptive allocation

Most multi-scale forecasters extract and combine temporal dependencies from the observed history to form hierarchical representations. N-HiTS couples multi-rate sampling with hierarchical synthesis, Pathformer selects pathways over input patch scales, and TimeMixer mixes decomposed histories with scale-specific predictors [? ? ?]. These methods define scale through input resolution, frequency structure or complete prediction branches. ISCF adopts an output-side perspective. Starting from one History State, it generates forecasts under multiple state-sharing scopes and allocates these granularities according to each future target.

Expert-based forecasters also use conditional allocation, but route among components with different modeling functions. MoLE mixes complete linear-centric forecasters, FreqMoE assigns frequency components to experts, and Time-MoE and Moirai-MoE activate sparse internal experts [? ? ? ?]. ISCF scopes are neither separately trained forecasters nor generic experts. They share the Encoder and Region-to-Step Forecast Generator, while Scope Projections construct information pools tied to explicit state-reuse extents. Scope Probabilities therefore allocate a sharing granularity for each target rather than selecting a model or frequency expert. BSCA jointly optimizes this scope-indexed forecast field and its allocation process. This output-side interpretation underlies the sharing-demand heterogeneity formalized in Section 3.

3. Problem Formulation and Empirical Motivation

The preceding discussion shows that varied-horizon forecasting involves more than sharing parameters across prediction lengths. A unified forecaster should return coherent predictions for future steps shared by different horizon requests, while its decoder must determine how predictive information is shared across the future domain. We now formalize these two issues and examine them through controlled empirical analyses, which motivate the decoder design developed in Section 4.

3.1. Varied-horizon forecasting and cross-horizon prefix consistency

Consider two forecasting requests with horizons $H_i < H_j$ issued from the same observed history. Their first H_i steps refer to identical future targets and should therefore receive identical predictions. We formalize this requirement below.

At forecast origin o, let

$$\mathbf{X}_o = [\mathbf{x}_{o-L+1}, \ldots, \mathbf{x}_o] \in \mathbb{R}^{L\times C}, \qquad \mathbf{Y}_o^{(H)} = [y_{o+\tau,c}]_{\tau=1,\ldots,H;\ c=1,\ldots,C} \in \mathbb{R}^{H\times C}.$$

Here, $\mathbf{X}_o$ is a length-L multivariate history, and $\mathbf{Y}_o^{(H)}$ contains the next H targets for C variables. Let $\mathcal{H} = \{H_1, \ldots, H_M\}$ denote the supported horizons and $T = \max \mathcal{H}$. The horizon H specifies the request endpoint, whereas τ indexes a particular future step. Thus, target (τ, c) is shared by all requests with $H \geq \tau$.

This shared-target relation defines **cross-horizon prefix consistency (CHPC)**. Let $\widehat{y}_{o+\tau,c}^{(H)}$ denote the prediction returned for target (τ, c) under a request with horizon H. Given identical history, forecast origin and preprocessing, CHPC requires

$$\widehat{y}_{o+\tau,c}^{(H_i)} = \widehat{y}_{o+\tau,c}^{(H_j)}, \qquad \forall\, H_i, H_j \in \mathcal{H},\ H_i < H_j, \quad 1 \leq \tau \leq H_i, \quad 1 \leq c \leq C.$$

Conventional long-term forecasting learns an independent predictor f_{θ_H} for each horizon. Because both parameters and optimization are horizon-specific, these predictors are not constrained to satisfy CHPC on their overlapping outputs.

A varied-horizon forecaster should instead use a single **future-step-indexed prediction function** $g_\theta(\mathbf{X}_o, \tau, c)$ and construct an H-step forecast as

$$\widehat{\mathbf{Y}}_o^{(H)} = [g_\theta(\mathbf{X}_o, \tau, c)]_{\tau=1,\ldots,H;\ c=1,\ldots,C}\,.$$

For a varied-horizon forecaster, all horizon requests query the same function at a shared target, so CHPC holds by construction. Changing the requested horizon alters only the returned prefix length, and each request becomes a nested view of one predicted trajectory.

3.2. Horizon-specific prefix inconsistency

Independently trained horizon-specific predictors may agree on some overlapping outputs, but their objectives do not enforce CHPC. We refer to a violation of CHPC as **cross-horizon prefix inconsistency** and quantify its magnitude through prediction disagreement on aligned histories and shared future targets.

At origin o, let $\widehat{\mathbf{Y}}_o^{(H_i)} \in \mathbb{R}^{H_i \times C}$ and $\widehat{\mathbf{Y}}_o^{(H_j)} \in \mathbb{R}^{H_j \times C}$ denote forecasts from independently trained models, where $H_i < H_j$. We define their **cross-horizon prefix disagreement (CHPD)** as

$$\mathrm{CHPD}_o(H_i, H_j) = \frac{1}{H_i C} \sum_{\tau=1}^{H_i} \sum_{c=1}^{C} \left| \widehat{y}_{o+\tau,c}^{(H_i)} - \widehat{y}_{o+\tau,c}^{(H_j)} \right| .$$

CHPD measures the mean absolute difference over the shared prefix in the original data scale. To aggregate variables with different magnitudes, we further define normalized CHPD (NCHPD):

$$\mathrm{NCHPD}(H_i, H_j) = \frac{1}{|\mathcal{O}|C} \sum_{o \in \mathcal{O}} \sum_{c=1}^{C} \frac{\frac{1}{H_i} \sum_{\tau=1}^{H_i} \left| \widehat{y}_{o+\tau,c}^{(H_i)} - \widehat{y}_{o+\tau,c}^{(H_j)} \right|}{\sigma_c^{\mathrm{train}} + \epsilon}.$$

Here, $\mathcal{O}$ denotes the aligned evaluation origins, $\sigma_c^{\mathrm{train}}$ is the train-split standard deviation of variable c, and $\epsilon > 0$ prevents division by zero. All comparisons use identical histories, forecast origins, scalers and overlapping target indices.

To visualize this inconsistency, we compare multi-horizon forecasts from DLinear models independently optimized on ETTh2. Fig. 2a shows that their predictions diverge over the same 96 future steps: relative to the $H = 720$ forecast, the forecasts for $H \in \{96, 192, 336\}$ differ by 2.51, 2.16 and 2.40 in mean absolute raw scale, respectively.

We further evaluate all 2,161 aligned ETTh2 validation origins and variables, with the resulting NCHPD matrix shown in Fig. 2b. NCHPD is non-zero for every horizon pair and is more pronounced between the longest and shorter requested horizons: the $H = 96$, $H = 192$ and $H = 336$ comparisons with $H = 720$ yield 0.0406, 0.0365 and 0.0366, whereas pairs among the three shorter horizons range from 0.0148 to 0.0166. These results show that the independently optimized DLinear models evaluated here fail to

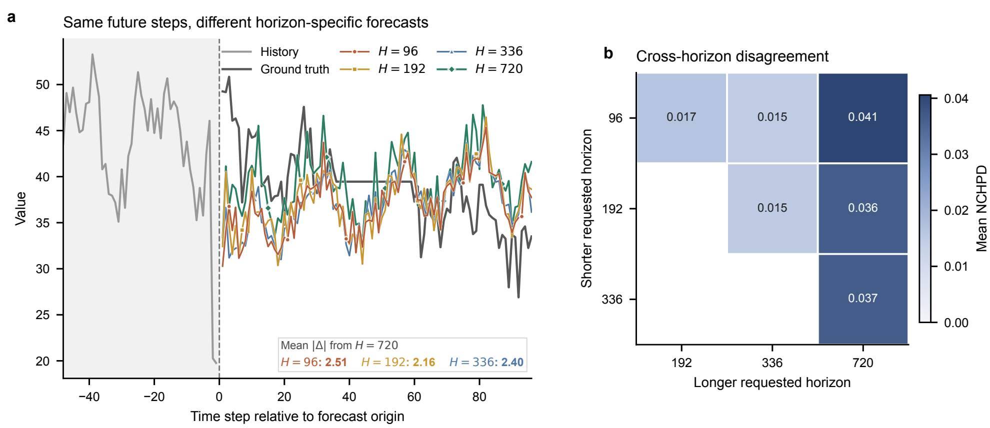

Figure 2: Independently optimized horizon-specific forecasts are inconsistent on shared future steps. **a**, An ETTh2 validation trajectory selected by aggregate CHPD, showing predictions from DLinear models trained separately for horizons 96, 192, 336 and 720. The panel includes 48 observed steps and the first 96 shared future steps; the inset reports mean absolute differences from the 720-step forecast. **b**, NCHPD averaged over all ETTh2 validation origins ($n = 2{,}161$) and variables.

form a prefix-consistent prediction trajectory, reflecting a structural limitation of horizon-specific forecasting: independently optimized models are not constrained to agree on shared future targets.

3.3. Future-region sharing-demand heterogeneity

CHPC specifies how predictions from different horizon requests should relate, but it does not determine how a unified decoder should construct the shared future trajectory. Simply adapting an architecture designed for one fixed horizon leaves this output-side requirement unresolved: a varied-horizon forecaster must learn from requests spanning short to long endpoints while producing one prefix-consistent trajectory. The decoder is therefore central to how predictive information is organized across the future domain, particularly in determining how broadly each history-conditioned state is reused before step-specific prediction.

We use the **sharing extent** s to denote the number of future steps that reuse one latent state. A **future region** $\mathcal{B}_b \subseteq \{1, \ldots, T\}$ is a contiguous subset of the maximum future domain rather than a requested horizon. Broad sharing can capture persistent trajectory structure but may smooth local variations, whereas fine sharing offers greater local flexibility with weaker cross-step regularization. We refer to variation in the preferred extent across regions as **future-region sharing-demand heterogeneity**.

To isolate this factor, we construct capacity-matched single-extent predictors that differ only in sharing extent; all other architecture, training and evaluation settings remain identical.

For aligned origin o, we define the **future-region prediction risk** of sharing extent s over region $\mathcal{B}_b$ as the empirical squared-error loss

$$R_{o,b,s} = \frac{1}{|\mathcal{B}_b|C} \sum_{\tau \in \mathcal{B}_b} \sum_{c=1}^{C} \left(\widehat{y}^{(s)}_{o+\tau,c} - y_{o+\tau,c} \right)^2 .$$

Operationally, $R_{o,b,s}$ is averaged over all future steps and variables in $\mathcal{B}_b$ for a given origin; it is an empirical region-level risk rather than an expectation over the data distribution. Lower $R_{o,b,s}$ indicates that extent s better matches region $\mathcal{B}_b$ within the controlled family. A region-dependent preference appears when $s^{\star}_{o,b} = \arg\min_s R_{o,b,s}$ changes with b.

We evaluate the five predictors with $s \in \{1, 8, 32, 128, 720\}$ across 12 contiguous 60-step future regions on ETTm2. Fig. 3a reports the percentage excess prediction risk of each extent above the lowest-risk extent within each region, with outlined squares marking the region-wise winners. The preferred extents vary across the future domain: each of the five extents wins two or three regions, all ten extent pairs exhibit bidirectional crossings beyond the predefined 0.5% margin, and the mean best-versus-second-best margin reaches 10.266%. No fixed extent minimizes prediction risk throughout this controlled example, supporting heterogeneous sharing demand across future regions.

Fig. 3b quantifies the performance upper bound associated with these region-wise preferences. Relative to the best fixed extent ($s = 720$), selecting the lowest-risk extent separately for each region reduces average MSE by 8.112%. Because the regional winners are selected using validation labels across separately trained predictors, this value represents descriptive oracle headroom rather than the realized gain of a learned decoder. Within this controlled example, its magnitude nevertheless shows that one fixed extent can leave meaningful region-specific headroom, motivating a decoder that can adapt sharing across the future domain.

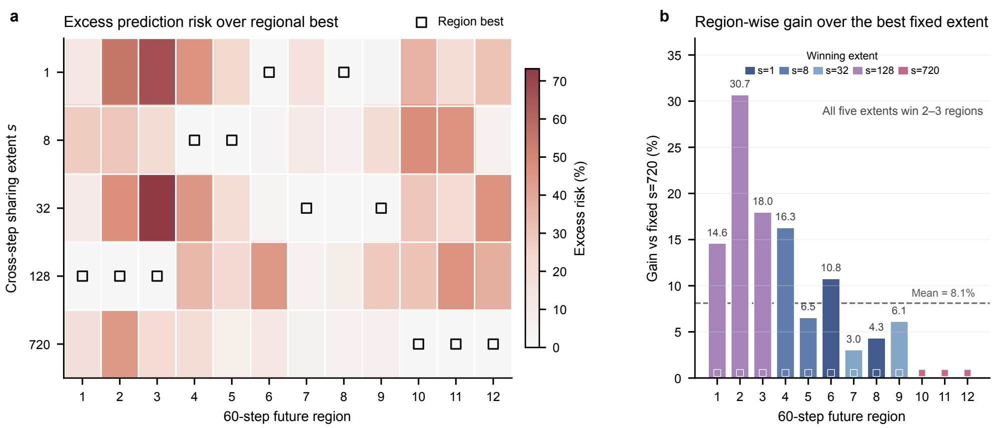

Figure 3: Preferred sharing extent varies across future regions. **a**, Percentage excess future-region prediction risk (mean squared error) of five capacity-matched single-extent predictors above the regional minimum for one selected ETTm2 validation example; outlined squares mark the best extent in each 60-step region. **b**, MSE reduction obtained by selecting the regional winner instead of the best fixed extent ($s = 720$); the dashed line denotes the 12-region mean.

4. ISCF-BSCA

Section 3 established two requirements for varied-horizon forecasting. Predictions for a shared future target should be invariant to the requested horizon, while the decoder should not impose one latent-state sharing extent on the entire future domain. We address these requirements with ISCF-BSCA. Independent Scope-Conditioned Forecasting (ISCF) is a decoder-side architecture that constructs one future-step-indexed trajectory from multiple sharing scopes, while Balanced Scope Co-Adaptation (BSCA) is the joint optimization method designed for it.

4.1. Architecture overview

Fig. 4 summarizes the forward computation of ISCF, which organizes its decoder into a **Scope Forecasting Path** and a **Target-Adaptive Allocation Path**. The architecture consumes a variable-wise History State rather than relying on a particular Encoder. Any Encoder that models temporal patch tokens and returns the required tensor interface can therefore serve as its backbone. This interface covers the patch-based Encoder family commonly used in time-series forecasting. Given a History Series $\mathbf{X} \in \mathbb{R}^{B\times L\times C}$, Patchify and the Encoder produce $\mathbf{R} \in \mathbb{R}^{B\times C\times R}$. The Scope Forecasting Path

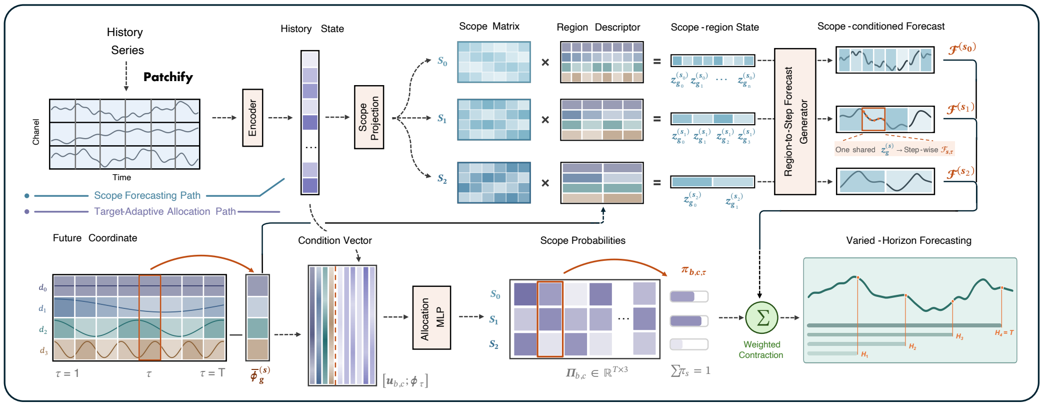

Figure 4: ISCF constructs one trajectory for Varied-Horizon Forecasting. The Scope Forecasting Path generates region-wise predictions under multiple sharing scopes. The Target-Adaptive Allocation Path assigns scope-conditioned information to each future target. Weighted contraction yields one trajectory whose nested prefixes answer different horizon requests. Three representative scopes are shown for clarity; the probability map and trajectories are schematic.

constructs region-wise predictions under each sharing scope and assembles them into $\mathcal{F}_\theta(\mathbf{X}) \in \mathbb{R}^{B\times C\times T\times S}$. The Target-Adaptive Allocation Path then determines how each future target draws on these sharing granularities.

The Target-Adaptive Allocation Path combines a projected History State with the Future Coordinate $\boldsymbol{\phi}_\tau$ and produces Scope Probabilities $\mathbf{\Pi} \in \mathbb{R}^{B\times C\times T\times S}$. Weighted contraction with the scope-indexed forecast field yields one trajectory $\widehat{\mathbf{Y}} \in \mathbb{R}^{B\times T\times C}$. For an H-step request, the region-local construction permits computation to be restricted to regions and targets intersecting the first H steps. The requested horizon changes only the evaluated prefix, not the computation assigned to a shared future target. ISCF therefore produces variable-length outputs while satisfying CHPC by construction.

The Scope-conditioned Forecasts and Scope Probabilities are optimized jointly. BSCA separates two training roles: the Uniform-Prefix Forecasting Loss optimizes varied-horizon prediction, while the Scope-Wise Forecasting Loss and Allocation-Balance Regularizer stabilize multi-scope learning. These objectives are described separately from the forward computation in Fig. 4.

4.2. History state and future coordinate

ISCF begins with the History State and Future Coordinate shown on the left of Fig. 4. The History Series is normalized per variable, divided into P

patches by Patchify and processed by the shared Encoder. This produces $\mathbf{Z} \in \mathbb{R}^{B\times C\times P\times D_e}$, which is flattened over its patch and embedding dimensions:

$$\mathbf{Z} = \mathcal{E}(\mathcal{N}(\mathbf{X}))\,, \qquad \mathbf{r}_{b,c} = \mathrm{vec}(\mathbf{Z}_{b,c}) \in \mathbb{R}^{R}, \qquad R = PD_e.$$

Collecting $\mathbf{r}_{b,c}$ over samples and variables gives the History State $\mathbf{R} = [\mathbf{r}_{b,c}] \in \mathbb{R}^{B\times C\times R}$. The preserved variable axis provides one history representation for each sample-variable pair, which is used by both decoder paths. ISCF requires only this tensor interface and does not otherwise constrain the internal design of the patch-token Encoder.

The History State summarizes the observed series, but a unified decoder must also identify where each prediction lies in the future domain. Using the requested horizon for this purpose would assign different conditioning contexts to the same target under different requests. ISCF instead introduces a **Future Coordinate** for every future step. This fixed coordinate supplies a horizon-independent target identity and a common positional basis for regions at different scopes. It also allows the Target-Adaptive Allocation Path to vary its scope preference across future steps.

Formally, the Future Coordinate is the field $\mathbf{\Phi} = [\boldsymbol{\phi}_1, \ldots, \boldsymbol{\phi}_T]^\top \in \mathbb{R}^{T\times D_q}$. We construct it from low-order discrete cosine functions, which provide smooth coordinate channels at progressively finer temporal frequencies. For $d = 0, \ldots, D_q - 1$, we first define

$$\widetilde{\phi}_{\tau,d} = \cos\left(\frac{\pi(\tau - \frac{1}{2})d}{T}\right).$$

The constant coordinate is $\phi_{\tau,0} = 1$. Each nonconstant coordinate is centered over the future domain and scaled as

$$\phi_{\tau,d} = \sqrt{2}\left(\widetilde{\phi}_{\tau,d} - \frac{1}{T}\sum_{j=1}^{T}\widetilde{\phi}_{j,d}\right), \qquad d \geq 1.$$

The constant channel preserves a global reference, while the centered nonconstant channels distinguish positions across the future domain. The resulting parameter-free field serves two roles. Region-wise averaging produces the Region Descriptors used to generate Scope-conditioned Forecasts, whereas the unpooled coordinate $\boldsymbol{\phi}_\tau$ identifies the individual target in the Condition Vector used for Scope Probabilities.

4.3. Generation of scope-conditioned forecasts

The Scope Forecasting Path contains S parallel scope lines, one for each sharing scope in $\mathcal{S} = \{s_1, \ldots, s_S\}$. A single Scope Projection stage contains an independently parameterized projection for every scope, so each line receives a dedicated Scope Matrix as its history-conditioned information pool. The matrix form is important because it retains a Future-Coordinate axis and a latent-mode axis. Region Descriptors can therefore query the same scope-specific history information at different future locations, without assigning a separate prediction head to every region.

For a scope s, the future domain is divided into contiguous regions

$$\mathcal{G}_g^{(s)} = \{(g-1)s + 1, \ldots, gs\}, \qquad g = 1, \ldots, T/s.$$

The Future Coordinate within each region is averaged to form the Region Descriptor:

$$\overline{\boldsymbol{\phi}}_g^{(s)} = \frac{1}{s} \sum_{\tau \in \mathcal{G}_g^{(s)}} \boldsymbol{\phi}_\tau \in \mathbb{R}^{D_q}.$$

For each scope, Scope Projection maps $\mathbf{r}_{b,c}$ into its dedicated Scope Matrix

$$\mathbf{M}_{b,c}^{(s)} = \mathrm{reshape}_{D_q \times K}\left(\mathbf{r}_{b,c}\mathbf{W}^{(s)} + \mathbf{b}^{(s)}\right) \in \mathbb{R}^{D_q \times K}.$$

The Region Descriptor selects and combines information from this matrix through a coordinate contraction, producing the region-wise Scope-region State

$$\mathbf{z}_{b,c,g}^{(s)} = \left(\overline{\boldsymbol{\phi}}_g^{(s)}\right)^\top \mathbf{M}_{b,c}^{(s)} \in \mathbb{R}^{K}.$$

Once the Scope Matrix is available, each Scope-region State depends only on the descriptor of its own region. Regions can therefore be evaluated separately and in parallel, although they draw on the same scope-specific history information. All future steps in $\mathcal{G}_g^{(s)}$ reuse $\mathbf{z}_{b,c,g}^{(s)}$. A finer scope constructs more region states and limits reuse to shorter intervals, whereas a broader scope shares each state over a longer interval. Scope thus defines a cross-step reuse pattern rather than a requested prediction horizon.

The Region-to-Step Forecast Generator converts each shared Scope-region State into step-wise predictions. Let $g_s(\tau)$ denote the region containing step

τ. The generator is shared across scopes and regions, and uses step-specific vectors $\mathbf{a}_\tau, \mathbf{n}_\tau \in \mathbb{R}^K$ and bias β_τ to define

$$\mathcal{F}_{b,c,\tau,s} = \left\langle \mathbf{a}_\tau, \mathbf{z}^{(s)}_{b,c,g_s(\tau)} \right\rangle + \left\langle \mathbf{n}_\tau, \mathrm{GELU}\left(\mathbf{z}^{(s)}_{b,c,g_s(\tau)}\right) \right\rangle + \beta_\tau.$$

Concatenating the separately generated regions gives the Scope-conditioned Forecast $\mathcal{F}^{(s)}$ for scope s. Collecting these forecasts produces the scope-indexed field $\mathcal{F}_\theta(\mathbf{X}) \in \mathbb{R}^{B\times C\times T\times S}$. Rather than ensembling separately trained models, ISCF jointly constructs this field through scope-specific projections and a shared Encoder and Region-to-Step Forecast Generator. This design couples forecast synthesis across scopes while avoiding duplicated encoder computation and generator parameters.

4.4. Target-adaptive scope allocation

Different future targets may require information organized at different sharing granularities. Estimating this preference requires both the dynamics encoded in the observed history and the target position in the future domain. The Target-Adaptive Allocation Path represents these two signals using a compact history summary and the Future Coordinate. It first projects the History State as

$$\mathbf{u}_{b,c} = \mathbf{W}_h \mathbf{r}_{b,c} + \mathbf{b}_h \in \mathbb{R}^{D_h}.$$

The compact summary $\mathbf{u}_{b,c}$ retains the sample-variable context used for allocation. For future step τ, the Condition Vector concatenates this summary with $\boldsymbol{\phi}_\tau$. The Allocation MLP maps the resulting vector to scope logits

$$\boldsymbol{\ell}_{b,c,\tau} = \mathbf{W}_o\, \mathrm{GELU}(\mathbf{W}_p[\mathbf{u}_{b,c}; \boldsymbol{\phi}_\tau] + \mathbf{b}_p) + \mathbf{b}_o \in \mathbb{R}^S,$$

$$\pi_{b,c,\tau,s} = \frac{\exp(\ell_{b,c,\tau,s})}{\sum_{s'=1}^{S} \exp(\ell_{b,c,\tau,s'})}.$$

Softmax converts these logits into the Scope Probability $\pi_{b,c,\tau,s}$ assigned to scope s. Collecting all probabilities gives $\boldsymbol{\Pi} \in \mathbb{R}^{B\times C\times T\times S}$, with $\sum_{s=1}^{S} \pi_{b,c,\tau,s} = 1$. Each probability vector parameterizes how strongly a future target draws on information organized at different sharing granularities. The final normalized prediction reads from the unified scope-indexed forecast field through weighted contraction:

$$\widehat{y}_{b,\tau,c} = \sum_{s=1}^{S} \pi_{b,c,\tau,s} \mathcal{F}_{b,c,\tau,s}.$$

This contraction is not an average over separately trained model outputs. The Scope Forecasting Path and Target-Adaptive Allocation Path are jointly learned within one decoder, allowing each target to receive a history-conditioned mixture of sharing granularities.

For an H-step request, ISCF needs only the Scope-region States whose regions intersect $1{:}H$ and the allocation entries for $\tau \leq H$. The resulting normalized prefix is transformed back to the original data scale and returned as

$$\widehat{\mathbf{Y}}_b^{(H)} = [\widehat{y}_{b,\tau,c}]_{\tau=1,\ldots,H;\ c=1,\ldots,C}.$$

Changing H restricts the active region and target computations but does not alter any prediction shared with a longer request. Every request is therefore a nested view of the same trajectory. The learned probabilities represent target-adaptive information allocation rather than an oracle scope selector. Later ablations and internal-behavior analyses evaluate whether this allocation yields useful scope differentiation.

4.5. Balanced Scope Co-Adaptation

Training ISCF introduces two coupled requirements. The fused trajectory requires uniform supervision across supported prefix lengths, while all scope lines must develop reliable forecasts as allocation is learned. BSCA therefore combines the **Uniform-Prefix Forecasting Loss** for varied-horizon prediction with two terms that stabilize multi-scope learning: the **Scope-Wise Forecasting Loss** and **Allocation-Balance Regularizer**.

Varied-horizon training should optimize every supported prediction length without privileging one endpoint. We therefore average raw-scale MAE equally over all prefixes $h = 1, \ldots, T$ to define the Uniform-Prefix Forecasting Loss. Before loss evaluation, inverse normalization maps the fused and Scope-conditioned Forecasts to the original data scale. Let $y_{b,\tau,c}$ denote the corresponding raw-scale target. The fused objective is

$$\mathcal{L}_{\text{prefix}} = \frac{1}{T} \sum_{h=1}^{T} \frac{1}{BCh} \sum_{b=1}^{B} \sum_{c=1}^{C} \sum_{\tau=1}^{h} \left| \widehat{y}_{b,\tau,c} - y_{b,\tau,c} \right|.$$

Exchanging the prefix and future-step summations gives an equivalent dense-prefix measure

$$\omega_\tau = \frac{1}{T}\sum_{h=\tau}^{T}\frac{1}{h}, \qquad \sum_{\tau=1}^{T}\omega_\tau = 1.$$

Under this measure, every prefix endpoint contributes equally, while a future step is weighted by the number of prefixes containing it. The same loss can therefore be written as

$$\mathcal{L}_{\text{prefix}} = \frac{1}{BC}\sum_{b=1}^{B}\sum_{c=1}^{C}\sum_{\tau=1}^{T}\omega_\tau \left|\widehat{y}_{b,\tau,c} - y_{b,\tau,c}\right|.$$

The Uniform-Prefix Forecasting Loss establishes the varied-horizon objective, but it does not resolve the joint optimization of multiple scope lines. Weighted contraction makes the fused-loss gradient received by each scope depend on its current Scope Probability. For a fixed target, this relation gives

$$\frac{\partial \mathcal{L}_{\text{prefix}}}{\partial \mathcal{F}_{b,c,\tau,s}} = \pi_{b,c,\tau,s}\frac{\partial \mathcal{L}_{\text{prefix}}}{\partial \widehat{y}_{b,\tau,c}}.$$

Consequently, early probability concentration can suppress learning in low-probability scope lines before their forecasts mature. The allocator then compares unevenly developed predictions, which can reinforce its initial preference. We first address this imbalance with the Scope-Wise Forecasting Loss, which applies the same prefix measure directly to every Scope-conditioned Forecast:

$$\mathcal{L}_{\text{scope}} = \frac{1}{BCS}\sum_{b=1}^{B}\sum_{c=1}^{C}\sum_{s=1}^{S}\sum_{\tau=1}^{T}\omega_\tau \left|\mathcal{F}_{b,c,\tau,s} - y_{b,\tau,c}\right|.$$

This term gives every scope a direct prediction gradient independent of its current Scope Probability. Direct supervision alone, however, does not regulate how the fused predictor distributes probability-mediated gradients. We therefore introduce the Allocation-Balance Regularizer using the uniform reference $q_s = 1/S$:

$$\mathcal{L}_{\text{balance}} = \frac{1}{BC}\sum_{b=1}^{B}\sum_{c=1}^{C}\sum_{\tau=1}^{T}\omega_\tau \frac{D_{\text{KL}}(\mathbf{q}\,\|\,\boldsymbol{\pi}_{b,c,\tau})}{\log S}.$$

The complete objective is

$$\mathcal{L}_{\text{BSCA}} = \mathcal{L}_{\text{prefix}} + \lambda_{\text{scope}}\mathcal{L}_{\text{scope}} + \lambda_{\text{balance}}(u)\mathcal{L}_{\text{balance}},$$

where $u \in [0, 1]$ is optimizer progress. In the frozen configuration, $\lambda_{\text{scope}} = 1$ and

$$\lambda_{\text{balance}}(u) = 0.1 \min\left(\frac{u}{0.25}, 1\right).$$

The regularizer penalizes strongly non-uniform Scope Probabilities, discouraging a small subset of scopes from dominating early optimization. Its ramp introduces this pressure gradually as the forecasting paths develop. This pressure promotes broader allocation and a more even distribution of probability-mediated gradients across scope lines. Together with direct scope supervision, it supports more balanced multi-scope optimization.

5. Experiments

ISCF-BSCA is designed to replace separately optimized horizon-specific predictors with one prefix-consistent forecaster. The central empirical question is whether a single model can serve all requested horizons without sacrificing forecasting accuracy. We examine this question through benchmark comparisons and deployment costs, and then use controlled ablations and validation diagnostics to identify the contribution and behavior of the proposed components.

5.1. Experimental setup

Datasets and metrics. We evaluate ISCF-BSCA on seven public multivariate forecasting benchmarks: ETTm1, ETTm2, ETTh1, ETTh2, Weather, Electricity (ECL) and Solar. Together, they cover electricity transformers, meteorology, electricity consumption and solar generation at different sampling frequencies and variable dimensions. Following the standard long-term forecasting protocol, we use prediction horizons $\mathcal{H} = \{96, 192, 336, 720\}$ and report mean squared error (MSE) and mean absolute error (MAE), where lower values indicate better forecasts. Further metric definitions, dataset descriptions, statistics, data splits and preprocessing details are provided in Appendix A.

Model and baselines. ISCF is a decoder framework that can be coupled with different Encoders. To focus the main experiments on the proposed decoder rather than a highly specialized history backbone, we pair ISCF with a lightweight patch-token MLP Encoder. To compare its forecasting performance against recent competitive methods, we select baselines from four dominant design families: (1) Transformer-based models, including SimpleTM, iTransformer, PatchTST and Crossformer; (2) lightweight linear or attention-based forecasters and training frameworks, including TimeAlign, QDF and DLinear; (3) convolutional models, including TVNet, ModernTCN and TimesNet; and (4) multi-scale or decomposition-based models, including AMD, TimeMixer and Leddam.

Evaluation protocols. We use two complementary comparisons. The first compares one ISCF-BSCA model per dataset with horizon-specific baselines that train a separate model for each requested horizon. It tests whether one unified predictor can replace four separately trained horizon-specific models while remaining competitive with horizon-wise optimization. The second evaluates every method under a one-model setting: each method uses one unified model trained for the maximum horizon, and an H-step request is evaluated from the first H outputs using the corresponding official fixed-H test loader.

Implementation details. Local experiments are implemented in Python 3.12.13 and PyTorch 2.9.0 with CUDA 12.8, and run on NVIDIA GeForce RTX 3090 GPUs. ISCF-BSCA is optimized with AdamW and a cosine learning-rate schedule; the learning rate, batch size, model width and training budget are configured per dataset. All locally reproduced baselines are built from their official codebases, and any source-informed configuration adaptation is documented explicitly.

5.2. Comparison with horizon-specific forecasters

The first comparison asks whether one unified forecaster can match baselines optimized separately for each requested horizon. ISCF-BSCA uses one model per dataset to produce all four forecasts, whereas each baseline entry in Table 1 follows its native horizon-specific protocol.

Table 1 reports the four-horizon mean for each dataset, while the complete horizon-wise results are provided in Appendix B. ISCF-BSCA obtains the best result in 13 of the 14 dataset–metric comparisons and the second-best result in the remaining comparison. Compared with TimeAlign, the strongest aggregate horizon-specific baseline, ISCF-BSCA uses one unified model instead of four separately trained horizon models and reduces the seven-dataset average MSE and MAE by 4.94% and 2.54%, respectively. The advantage persists

Dataset	ISCF-BSCA (Ours)	TimeAlign	QDF	AMD	SimpleTM	TVNet	iTransformer	TimeMixer	Leddam	ModernTCN	PatchTST	Crossformer	TimesNet	DLinear
ETTm1	**0.329**/**0.363**	0.341/0.368	0.353/0.380	0.351/0.378	0.383/0.396	0.348/0.379	0.362/0.391	0.356/0.380	0.360/0.390	0.351/0.381	0.353/0.376	0.416/0.435	0.399/0.406	0.356/0.378
ETTm2	0.248/**0.304**	**0.245**/**0.304**	0.257/0.315	0.254/0.315	0.280/0.325	0.251/0.311	0.269/0.329	0.257/0.318	0.265/0.321	0.253/0.314	0.256/0.317	0.518/0.501	0.292/0.331	0.259/0.324
ETTh1	**0.390**/**0.415**	0.418/0.431	0.439/0.442	0.412/0.428	0.428/0.433	0.407/0.421	0.439/0.448	0.427/0.441	0.415/0.431	0.404/0.420	0.419/0.437	0.440/0.463	0.488/0.452	0.425/0.439
ETTh2	**0.306**/**0.364**	0.347/0.387	0.353/0.394	0.366/0.406	0.356/0.392	0.324/0.377	0.374/0.406	0.349/0.397	0.346/0.392	0.322/0.379	0.351/0.395	0.809/0.658	0.390/0.417	0.431/0.447
Weather	**0.214**/**0.245**	0.218/0.247	0.237/0.271	0.225/0.265	0.243/0.271	0.221/0.261	0.233/0.271	0.226/0.264	0.227/0.264	0.224/0.264	0.226/0.264	0.228/0.287	0.259/0.287	0.242/0.293
ECL	**0.151**/**0.244**	0.156/0.246	0.167/0.260	0.162/0.257	0.168/0.262	0.165/0.254	0.164/0.261	0.173/0.272	0.162/0.257	0.156/0.253	0.159/0.253	0.181/0.279	0.193/0.295	0.166/0.264
Solar	**0.188**/**0.208**	0.196/0.217	0.208/0.258	0.205/0.248	0.191/0.251	0.228/0.277	0.202/0.248	0.193/0.252	0.224/0.265	0.228/0.282	0.194/0.245	0.205/0.232	0.244/0.334	0.223/0.226
Average	**0.261**/**0.306**	0.274/0.314	0.288/0.331	0.282/0.328	0.293/0.333	0.278/0.326	0.292/0.336	0.283/0.332	0.286/0.331	0.277/0.327	0.280/0.326	0.399/0.408	0.324/0.360	0.300/0.339

Table 1: Comparison with horizon-specific forecasters. Each dataset entry reports MSE/MAE averaged over $H \in \{96, 192, 336, 720\}$, and the final row is the unweighted mean over seven datasets. ISCF-BSCA uses one unified model per dataset, whereas each baseline follows its horizon-specific protocol. Full horizon-wise results, source roles and protocol details are provided in Appendix B. Best and second-best displayed values are marked in red bold and blue underline, respectively.

across datasets with markedly different dimensionalities. Over the four low-dimensional ETT benchmarks, ISCF-BSCA improves the average MSE and MAE over TimeAlign by 5.69% and 2.89%; on the high-dimensional ECL and Solar datasets, the corresponding improvements are 3.90% and 2.27%. Notably, these results are obtained with a lightweight patch-token MLP Encoder rather than a specialized high-capacity backbone, indicating that the ISCF decoder can construct accurate future trajectories from comparatively shallow history representations. Overall, the results show that ISCF-BSCA improves forecasting accuracy over the evaluated horizon-specific baselines while consolidating the four requested horizons into one model.

5.3. One-model-all-horizons evaluation

The preceding comparison showed that one unified ISCF-BSCA model can outperform baselines that train a separate model for each of the four horizons. We next place every method under the same one-model-all-horizons workflow. For each dataset, a single model produces the maximum-length forecast, and requests for $H \in \{96, 192, 336\}$ are evaluated from the corresponding output prefixes without retraining or reselection. This setting maintains consistency across overlapping predictions within each method and provides a more deployment-relevant comparison without horizon-specific model specialization.

Table 2 summarizes the four-horizon mean for each dataset, with complete horizon-wise results provided in Appendix B. Under the shared one-model constraint, ISCF-BSCA ranks first in all 14 dataset–metric comparisons. Averaged over all seven datasets and four horizons, it reduces MSE and MAE by 6.45% and 3.72% relative to TimeAlign, the next strongest complete baseline in Table 2. These gains exceed the corresponding 4.94% and 2.54% improvements in the horizon-specific comparison by 1.51 and 1.18 percentage

Dataset	ISCF-BSCA (Ours)	TimeAlign	QDF	AMD	SimpleTM	iTransformer	PatchTST	DLinear
ETTm1	**0.329**/**0.363**	0.345/0.371	0.352/0.384	0.354/0.380	0.384/0.399	0.409/0.415	0.354/0.384	0.359/0.381
ETTm2	**0.248**/**0.304**	0.250/0.308	0.257/0.316	0.255/0.316	0.285/0.330	0.294/0.339	0.258/0.316	0.292/0.359
ETTh1	**0.390**/**0.415**	0.420/0.433	0.452/0.453	0.416/0.430	0.434/0.436	0.468/0.458	0.429/0.440	0.474/0.480
ETTh2	**0.306**/**0.364**	0.368/0.399	0.343/0.392	0.370/0.411	0.356/0.393	0.385/0.408	0.352/0.395	0.415/0.439
Weather	**0.214**/**0.245**	0.220/0.250	0.260/0.295	0.226/0.266	0.254/0.280	0.262/0.284	0.231/0.268	0.247/0.301
ECL	**0.151**/**0.244**	0.157/0.246	0.183/0.281	0.163/0.257	0.169/0.265	0.177/0.269	0.161/0.255	0.167/0.265
Solar	**0.188**/**0.208**	0.193/0.219	0.207/0.268	0.204/0.250	0.190/0.247	0.229/0.261	0.194/0.247	0.253/0.313
Average	**0.261**/**0.306**	0.279/0.318	0.293/0.341	0.284/0.330	0.296/0.336	0.318/0.348	0.283/0.329	0.315/0.363

Table 2: One-model-all-horizons comparison. Each method uses one unified model per dataset, and each entry reports MSE/MAE averaged over $H \in \{96, 192, 336, 720\}$. The final row is the unweighted mean over seven datasets. Full horizon-wise results and protocol details are provided in Appendix B. Best and second-best displayed values are marked in red bold and blue underline, respectively.

points, respectively, showing that the advantage of ISCF-BSCA becomes more pronounced when every method follows the same unified forecasting workflow.

These results establish ISCF-BSCA as an effective forecaster for unified multi-horizon prediction. Its scope-indexed forecast field constructs one prefix-consistent trajectory, while Target-Adaptive Scope Allocation integrates predictive information across different sharing granularities for each future step. The consistently strong results across the four evaluated horizons support this output-side design for forecasting requests ranging from short to long endpoints.

5.4. Accuracy and system cost

Forecast accuracy is only one consideration in practical forecasting services. Runtime memory and model storage also determine deployment and management costs, particularly in resource-constrained settings. We therefore measure the peak allocated GPU memory of the complete inference service and the total checkpoint storage required to serve all four horizons. Fig. 5 compares forecasting accuracy with these two costs. ISCF-BSCA uses one unified checkpoint per dataset, whereas each baseline retains four horizon-specific model instances.

As shown in Fig. 5, ISCF-BSCA attains the lowest macro MSE among the eight displayed methods, at 0.261, while requiring 17.677 MiB of checkpoint storage and 38.817 MiB of peak allocated inference memory. Compared with TimeAlign, AMD, iTransformer, PatchTST and TimeMixer, the unified model reduces checkpoint storage by 30.12–92.01% and peak memory by 18.01–83.69%. Relative to TimeAlign in particular, ISCF-BSCA lowers MSE by 4.94%, checkpoint storage by 81.48% and peak memory by 64.42%.

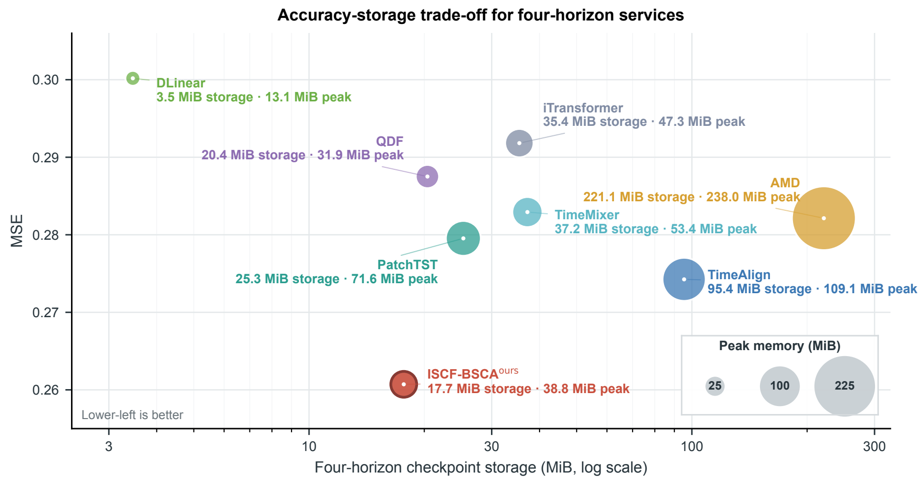

Figure 5: Accuracy and system cost for four-horizon forecasting. Macro MSE is plotted against total checkpoint storage, and bubble area represents peak allocated inference memory. ISCF-BSCA uses one unified checkpoint, whereas each baseline uses four horizon-specific models. The storage axis is logarithmic; lower-left positions indicate a more favorable trade-off.

By consolidating four forecasting horizons into one model, ISCF-BSCA achieves a favorable balance between accuracy, runtime memory and checkpoint storage. It is not the lightest method on every resource dimension, as DLinear and QDF retain lower values on at least one cost measure, but it combines the best forecasting accuracy with substantially lower resource footprints than several competitive baselines. This balance makes the unified architecture attractive for practical forecasting services in which predictive quality and model-management cost must be considered together.

5.5. Component and training-objective ablations

The preceding experiments establish the overall effectiveness of ISCF-BSCA. To determine whether its individual components contribute to this performance, we conduct strictly controlled ablations in which each variant is trained end to end under the same protocol. **w/o BSCA** retains the Uniform-Prefix Forecasting Loss but removes the Scope-Wise Forecasting Loss and Allocation-Balance Regularizer. **w/o Target-Adaptive Allocation** replaces learned Scope Probabilities with equal non-adaptive fusion. **Shared Scope Projection** replaces the scope-specific history projections with one shared

Table 3: Core ablation of ISCF-BSCA. Each dataset entry is the mean over prediction horizons $H \in \{96, 192, 336, 720\}$. Lower is better. Best and second-best results are shown in bold and underlined, respectively.

Variant	ETTm1		ETTm2		ETTh1		ETTh2		Weather		Avg	
	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
Full ISCF-BSCA	**0.342**	**0.365**	**0.253**	**0.309**	**0.407**	**0.432**	**0.311**	**0.367**	**0.214**	**0.245**	**0.305**	**0.344**
w/o BSCA	0.353	0.378	<u>0.258</u>	0.315	0.434	0.450	0.319	0.375	0.218	0.250	0.316	0.354
w/o Target-Adaptive Allocation	<u>0.347</u>	<u>0.369</u>	0.259	0.317	<u>0.413</u>	0.437	<u>0.315</u>	<u>0.371</u>	<u>0.217</u>	<u>0.249</u>	<u>0.310</u>	<u>0.349</u>
Shared Scope Projection	0.355	0.376	<u>0.258</u>	<u>0.314</u>	0.415	<u>0.435</u>	0.325	0.382	<u>0.217</u>	0.250	0.314	0.351
Fixed Scope ($s = 144$)	0.351	0.372	0.263	0.319	0.428	0.441	0.318	0.373	<u>0.217</u>	<u>0.249</u>	0.315	0.351

projection, whereas **Fixed Scope** ($s = 144$) retains only the preregistered middle scope.

As reported in Table 3, Full ISCF-BSCA achieves a five-dataset macro MSE/MAE of 0.305/0.344 and ranks first in all 12 dataset and average metric columns. Removing BSCA produces the largest aggregate degradation, increasing MSE and MAE by 3.48% and 2.83%. Equal fusion yields 0.310/0.349, corresponding to gains of 1.61% in MSE and 1.43% in MAE from Target-Adaptive Allocation. Shared Scope Projection and Fixed Scope yield MSE values of 0.314 and 0.315, respectively, compared with 0.305 for the complete model. Full ISCF-BSCA improves both metrics over every control on all five datasets.

Together, these controlled comparisons show that each evaluated component contributes to the complete model. BSCA improves joint optimization across the scope lines. Target-Adaptive Scope Allocation outperforms equal fusion, while scope-specific projections and multi-scope generation improve upon their shared-projection and fixed-scope counterparts. Their combination yields the strongest overall performance.

5.6. Scope diversity and allocation behavior

The ablation results establish the contribution of the individual ISCF-BSCA components to forecasting accuracy. We next examine whether the internal behavior of the complete model follows the intended design. Specifically, we ask whether the jointly trained scope lines retain distinct scope-dependent forecast signals and whether the Target-Adaptive Allocation Path produces diverse scope preferences across future regions. Together, these properties determine whether ISCF can synthesize region-specific forecasts from multiple sharing granularities rather than collapsing to repeated scope trajectories and a fixed allocation pattern.

We first extract the region-wise predictions from each scope line and concatenate them into a complete Scope-conditioned Forecast. Directly overlaying the five trajectories obscures their differences, so Fig. 6b visualizes each Scope-conditioned Forecast relative to the fused trajectory, $\mathcal{F}_{\tau}^{(s)} - \widehat{y}_{\tau}$. The resulting profiles exhibit distinct temporal structures, with a mean pairwise absolute disagreement of 0.0812 on the normalized scale. The scope-indexed forecast field therefore retains distinct candidate signals for region-wise integration rather than collapsing to five repeated trajectories.

We next examine how the Target-Adaptive Allocation Path combines these heterogeneous scope signals. Fig. 6a colours the fused maximum-horizon forecast by the scope receiving the highest mean Scope Probability within each future region. All five scopes become the highest-weight component in at least one of the eight regions, showing that the allocation does not reduce to one fixed regional preference. Fig. 6c further displays the complete per-step Scope Probabilities, whose relative ordering varies across future steps and regions rather than remaining identical throughout the future domain. Notably, the farthest region assigns its highest mean weight to the largest sharing scope, $s = 720$. This sample-level observation is consistent with the intuition that distal forecasting can benefit from broader information sharing. Together, the three panels show how distinct scope signals and region-dependent soft allocation support region-specific forecast synthesis. These behaviors operate within one shared prediction trajectory: prefix-based inference guarantees CHPC, and the frozen implementation yields a maximum absolute CHPD of 0 in all 20 dataset–horizon checks.

5.7. Generalization studies

ISCF is designed as an Encoder-agnostic plug-in decoder: it operates on an Encoder-produced history representation rather than relying on a particular history-modeling architecture. We examine this versatility using DLinear and PatchTST as representatives of lightweight linear and Transformer-based Encoders, respectively. For each backbone, we replace the Original Decoder with the complete ISCF-BSCA framework and train the resulting model end to end. The comparison covers Weather, ETTm1 and ETTm2, and each displayed value averages MSE over the four horizons.

As shown in Fig. 7, replacing the Original Decoder with ISCF-BSCA improves forecasting performance on both example backbones. For DLinear, the replacement lowers MSE on all three datasets and reduces the macro average from 0.303 to 0.286, corresponding to an improvement of 5.61%. PatchTST shows smaller but consistent gains, with macro MSE decreasing from 0.282 to 0.276, corresponding to an improvement of 2.13%. The improvements obtained with both a lightweight linear backbone and a Transformer-based backbone show that ISCF-BSCA is not tied to a single Encoder design, supporting its intended role as a general plug-in decoder for unified multi-horizon forecasting.

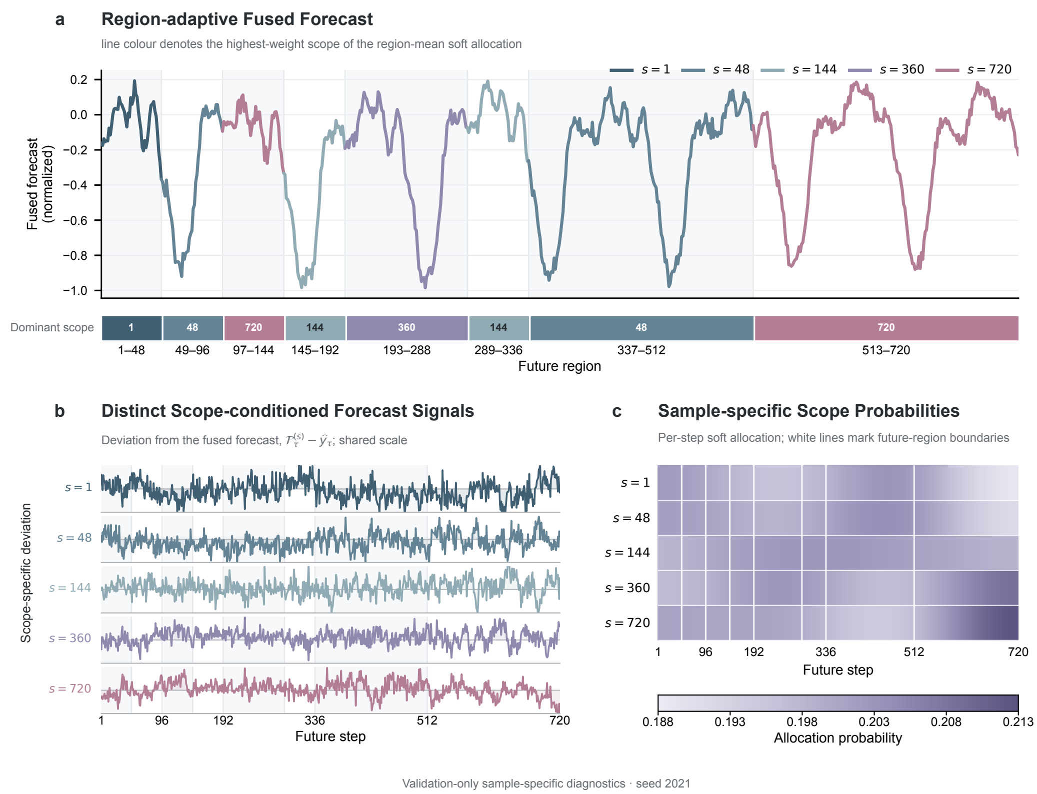

Figure 6: Scope diversity and sample-specific allocation behavior in ISCF-BSCA. All panels use ETTm1 validation probe 113. The probe was selected from 1,280 candidates by first maximizing regional dominant-scope diversity and then maximizing mean pairwise scope-forecast disagreement. **a**, The actual fused forecast; line colour and the lower strip identify the highest-weight scope after averaging Scope Probabilities within each future region. This encoding summarizes a soft allocation rather than a hard route. **b**, Deviations of the five Scope-conditioned Forecasts from the fused forecast on a shared scale. **c**, Per-step Scope Probabilities for the same probe; white lines mark future-region boundaries. The panels are validation-only diagnostics from seed 2021 and illustrate a selected example rather than population prevalence, oracle scope recovery or causal specialization.

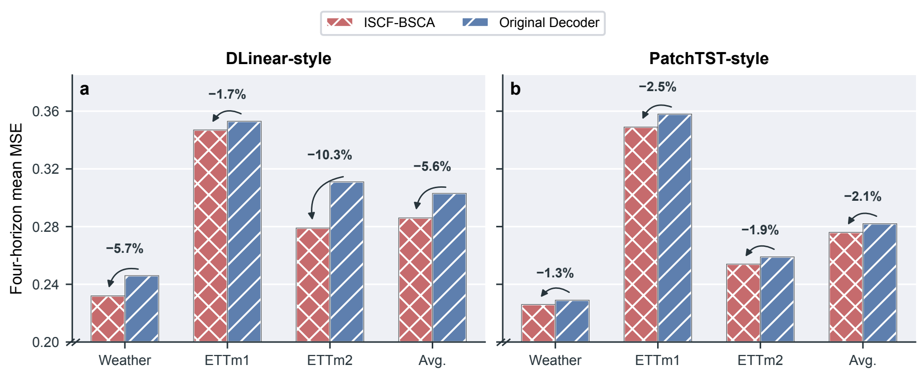

Figure 7: Generalization across two forecasting backbones. Four-horizon mean MSE is reported for **a**, DLinear and **b**, PatchTST with either the Original Decoder or ISCF-BSCA. Percentages denote the relative MSE change, and Avg. is the mean over Weather, ETTm1 and ETTm2. The MSE axis begins at 0.20 to resolve paired differences.

6. Discussion and Limitations

Varied-horizon forecasting requires one forecasting system to serve different request endpoints from a shared prediction trajectory. Horizon-specific predictors optimize each request endpoint separately and therefore do not impose CHPC. ISCF instead maps a shared history representation and future-step coordinates to one prediction trajectory, so requests with different endpoints become nested prefixes of the same forecast. The experiments in Sections 5.2 and 5.3 show that this consistency constraint can coexist with competitive accuracy, while Fig. 5 shows the deployment benefit of consolidating the horizon service into one unified checkpoint.

Unlike many existing multi-scale encoder studies, which extract and mix historical dependencies at multiple input resolutions, ISCF starts from the output side and organizes how one shared representation generates the future. In these encoders, scale refers to the granularity of historical representation before decoding [? ? ?]. ISCF uses a different axis: it generates candidate futures whose latent states are reused over different contiguous extents. A scope therefore indexes output-side sharing granularity rather than input resolution or receptive field. This distinction shifts the multi-scale question from representation learning to forecast generation. The scope-wise ablations, the future-region analysis in Section 3 and the Fig. 6 diagnostics together suggest that different future regions can benefit from different sharing extents, while Target-Adaptive Allocation assigns step-specific preferences among these candidate signals. Multi-scale encoders and ISCF are therefore methodologically distinct yet complementary: multi-scale encoders organize how historical evidence is represented, whereas ISCF organizes how a shared representation generates forecasts with multiple future-sharing granularities.

Several methodological limitations delimit the current formulation. First, ISCF uses a finite set of pre-specified contiguous scopes and future regions. This design assumes that useful sharing patterns can be described by fixed intervals; processes whose dependencies change at irregular locations may require data-adaptive region partitioning. Second, each future region is generated from its own scope-conditioned state, while relationships between regions are represented through the shared history state and future coordinates rather than a dedicated mechanism that passes information from one predicted region to another. Finally, retaining multiple scope branches increases decoder-side computation relative to a single-scope head, even though shared encoders and unified serving reduce duplicated computation and storage. These limitations motivate future extensions with adaptive region partitioning and explicit cross-region forecasting interactions.

7. Conclusion

Varied-horizon forecasting requires one model to serve different request endpoints without changing predictions for their shared future steps. We formulated this requirement as cross-horizon prefix consistency (CHPC) and identified future-region sharing-demand heterogeneity as a complementary decoder-side challenge. To address both, ISCF constructs a scope-indexed forecast field through region-wise generation at multiple output-side sharing extents and integrates its Scope-conditioned Forecasts through Target-Adaptive Allocation. BSCA supports the joint learning of the fused trajectory and all scope lines without changing the inference graph; the resulting horizon-agnostic trajectory provides the nested prefixes returned for different horizon requests.

Across seven multivariate benchmarks and four forecast horizons, ISCF-BSCA used one unified model per dataset, achieved stronger aggregate forecasting accuracy than the evaluated horizon-specific baselines and ranked first under the one-model-all-horizons comparison. The unified architecture also achieved a favorable balance between forecasting accuracy, checkpoint storage and inference memory. Controlled ablations supported the contribution of multi-scope generation, scope-specific projections, Target-Adaptive Allocation and BSCA. A selected internal diagnostic illustrated distinct scope signals and region-dependent allocation, while experiments with DLinear and PatchTST supported decoder compatibility across the two evaluated Encoder families. Together, these results support output-side sharing granularity as a practical design axis for unified forecasters that serve multiple horizons through one prefix-consistent trajectory.

References

Appendix A. Experiment Details

Appendix A.1. Metric Details

We use mean squared error (MSE) and mean absolute error (MAE) to evaluate forecasting accuracy. Given the ground-truth values y_i and their predictions $\widehat{y}_i$, the two metrics are defined as

$$\mathrm{MSE} = \frac{1}{N}\sum_{i=1}^{N}\left(y_i - \widehat{y}_i\right)^2, \qquad \mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}\left|y_i - \widehat{y}_i\right|,$$

where N is the total number of scalar predictions over all evaluation windows, forecast steps and variables for the requested horizon. Lower MSE and MAE indicate more accurate forecasts.

Appendix A.2. Datasets

We evaluate ISCF-BSCA on seven widely used multivariate time-series forecasting datasets spanning electricity transformers, electricity consumption, meteorology and solar generation. These benchmarks differ substantially in variable dimension, sampling frequency and temporal dynamics. Table A.1 summarizes their basic statistics.

1. **ETT (Electricity Transformer Temperature)** records seven load and temperature factors from two electricity transformers in two regions of China between 2016 and 2018 [**?**]. We use its four standard subsets: ETTh1 and ETTh2 are sampled hourly, whereas ETTm1 and ETTm2 are sampled every 15 minutes.
2. **Electricity (ECL)** records the hourly electricity consumption of 321 clients from 2012 to 2014 and is provided by the UCI Machine Learning Repository [**?**].
3. **Weather** is collected by the Beutenberg Weather Station at the Max Planck Institute for Biogeochemistry in Jena, Germany. It contains 21 meteorological indicators sampled every 10 minutes during 2020 [**?**].
4. **Solar** contains solar-power measurements collected every 10 minutes from 137 photovoltaic plants in Alabama during 2006 [**?**].

Dataset Size is reported as (Train, Validation, Test). The ETT datasets follow their standard fixed chronological splits, whereas Weather, ECL and Solar use chronological 70/10/20 splits. The scaler is fitted exclusively on the training segment. All datasets are evaluated at prediction lengths $H \in \{96, 192, 336, 720\}$.

Appendix A.3. Implementation Details

The local experiments are implemented in Python 3.12.13 and PyTorch 2.9.0 with CUDA 12.8. Each training run uses one NVIDIA GeForce RTX

Table A.1: Dataset Statistics.

Dataset	Variables	Sampling Frequency	Dataset Size	Domain
ETTh1	7	1 h	(8545, 2881, 2881)	Electricity
ETTh2	7	1 h	(8545, 2881, 2881)	Electricity
ETTm1	7	15 min	(34465, 11521, 11521)	Electricity
ETTm2	7	15 min	(34465, 11521, 11521)	Electricity
ECL	321	1 h	(18317, 2633, 5261)	Electricity
Weather	21	10 min	(36792, 5271, 10540)	Weather
Solar	137	10 min	(36601, 5161, 10417)	Solar energy

Table A.2: Training and Evaluation Settings.

Dataset	Max Length	Initial LR	Batch	Grad. Accum.	Max epochs	Patience
ETTh1	720	3×10^{-4}	32	1	45	10
ETTh2	720	5×10^{-4}	32	1	30	7
ETTm1	720	1×10^{-4}	16	2	30	7
ETTm2	720	5×10^{-5}	16	2	60	12
Weather	720	2×10^{-5}	32	1	120	24
ECL	720	5×10^{-4}	4	8	30	7
Solar	720	3×10^{-4}	16	2	45	10

3090 GPU. ISCF-BSCA is trained from scratch with AdamW and a cosine learning-rate schedule. Table A.2 reports the dataset-specific optimization settings, and Table A.3 lists the corresponding Encoder interface and ISCF-BSCA configuration.

Gradient Accumulation Steps denotes the number of mini-batches whose gradients are accumulated before one optimizer update; the effective batch size is therefore the batch size multiplied by this value. All runs use seed 2021. The checkpoint with the lowest validation mean MSE over $H \in \{96, 192, 336, 720\}$ is retained and serves all four forecasting requests. Weight decay is 0.01 except for ETTm2, where it is 0.001, and early stopping uses zero minimum improvement.

The scope set and BSCA coefficients are shared across datasets. Specifically, $\lambda_{\text{scope}} = 1$ and $\lambda_{\text{balance}}(u) = 0.1 \min(u/0.25, 1)$, where $u \in [0, 1]$ denotes normalized optimizer progress.

Table A.3: ISCF-BSCA Configuration.

Dataset	Scope set	Mode rank	Patch number	d_{model}	d_{ff}	Dropout	Weight decay	λ_{scope}	$\lambda_{\text{balance}}^{\text{max}}$ / ramp
ETTh1	$\{1, 48, 144, 360, 720\}$	109	24	32	32	0.1	0.01	1.0	0.1 / 0.25
ETTh2	$\{1, 48, 144, 360, 720\}$	116	12	64	128	0.1	0.01	1.0	0.1 / 0.25
ETTm1	$\{1, 48, 144, 360, 720\}$	116	1	128	256	0.9	0.01	1.0	0.1 / 0.25
ETTm2	$\{1, 48, 144, 360, 720\}$	64	6	128	128	0.2	0.001	1.0	0.1 / 0.25
Weather	$\{1, 48, 144, 360, 720\}$	116	19	64	128	0.0	0.01	1.0	0.1 / 0.25
ECL	$\{1, 48, 144, 360, 720\}$	64	1	256	1,024	0.3	0.01	1.0	0.1 / 0.25
Solar	$\{1, 48, 144, 360, 720\}$	128	4	256	256	0.3	0.01	1.0	0.1 / 0.25

Appendix B. Full Results

Tables B.1 and B.2 provide the complete results for the two forecasting comparisons in Sections 5.2 and 5.3. The main text reports the four-horizon average for each dataset, whereas the appendix retains MSE and MAE at every evaluated horizon. Table B.1 compares one unified ISCF-BSCA model with baselines optimized separately for each requested horizon. Table B.2 places every method under the same one-model-all-horizons workflow, where shorter requests are evaluated from the corresponding prefixes of one maximum-length forecast.

Dataset	H	ISCF-BSCA (Ours)		TimeAlign (2026)		QDF (2026)		AMD (2025)		SimpleTM (2025)		TVNet (2025)		iTransformer (2024b)		TimeMixer (2024)		Leddam (2024)		ModernTCN (2024)		PatchTST (2023)		Crossformer (2023)		TimesNet (2023)		DLinear (2023)	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTm1	96	**0.265**	**0.326**	0.280	0.331	0.283	0.336	0.289	0.343	0.325	0.363	0.288	0.343	0.300	0.353	0.293	0.345	0.294	0.347	0.292	0.346	0.293	0.346	0.310	0.361	0.338	0.375	0.300	0.345
	192	**0.304**	**0.349**	0.323	0.357	0.336	0.369	0.329	0.366	0.361	0.381	0.326	0.367	0.345	0.382	0.335	0.372	0.334	0.370	0.332	0.368	0.333	0.370	0.363	0.402	0.371	0.387	0.336	0.366
	336	**0.343**	0.371	0.352	0.376	0.369	0.390	0.365	0.386	0.391	0.404	0.365	0.391	0.374	0.398	0.368	0.386	0.392	0.425	0.365	0.391	0.369	**0.369**	0.389	0.430	0.410	0.411	0.367	0.386
	720	**0.404**	**0.405**	0.408	0.409	0.423	0.424	0.423	0.417	0.454	0.438	0.412	0.413	0.429	0.430	0.426	0.417	0.421	0.419	0.416	0.417	0.416	0.420	0.600	0.547	0.478	0.450	0.419	0.416
	Avg.	**0.329**	**0.363**	0.341	0.368	0.353	0.380	0.351	0.378	0.383	0.396	0.348	0.379	0.362	0.391	0.356	0.380	0.360	0.390	0.351	0.381	0.353	0.376	0.416	0.435	0.399	0.406	0.356	0.378
ETTm2	96	0.159	**0.242**	**0.157**	0.244	0.164	0.253	0.168	0.258	0.175	0.258	0.161	0.254	0.175	0.266	0.165	0.256	0.174	0.260	0.166	0.256	0.166	0.256	0.263	0.359	0.189	0.265	0.164	0.255
	192	0.218	0.285	**0.210**	**0.280**	0.218	0.291	0.221	0.295	0.240	0.301	0.220	0.293	0.242	0.312	0.225	0.298	0.231	0.301	0.222	0.293	0.223	0.296	0.345	0.400	0.254	0.310	0.224	0.304
	336	0.270	0.318	**0.265**	0.318	0.282	0.330	0.270	0.327	0.305	0.342	0.272	**0.316**	0.282	0.340	0.277	0.332	0.288	0.336	0.272	0.324	0.274	0.329	0.469	0.496	0.313	0.345	0.277	0.337
	720	**0.343**	**0.371**	0.346	0.374	0.364	0.385	0.355	0.381	0.401	0.399	0.349	0.379	0.378	0.398	0.360	0.387	0.368	0.386	0.351	0.381	0.362	0.385	0.996	0.750	0.413	0.402	0.371	0.401
	Avg.	0.248	**0.304**	**0.245**	**0.304**	0.257	0.315	0.254	0.315	0.280	0.325	0.251	0.311	0.269	0.329	0.257	0.318	0.265	0.321	0.253	0.314	0.256	0.317	0.518	0.501	0.292	0.331	0.259	0.324
ETTh1	96	**0.348**	**0.386**	0.380	0.407	0.370	0.394	0.371	0.399	0.367	0.393	0.371	0.408	0.386	0.405	0.372	0.401	0.377	0.394	0.368	0.394	0.377	0.397	0.386	0.426	0.384	0.402	0.379	0.403
	192	**0.377**	**0.403**	0.406	0.419	0.422	0.427	0.403	0.420	0.429	0.427	0.398	0.409	0.424	0.440	0.413	0.430	0.408	0.427	0.405	0.413	0.409	0.428	0.413	0.442	0.557	0.436	0.408	0.419
	336	0.393	0.414	0.427	0.432	0.464	0.455	0.423	0.432	0.447	0.445	0.401	**0.409**	0.449	0.460	0.438	0.450	0.424	0.437	**0.391**	0.412	0.431	0.444	0.440	0.461	0.491	0.469	0.440	0.440
	720	**0.443**	**0.458**	0.459	0.464	0.499	0.493	0.452	0.461	0.470	0.465	0.458	0.459	0.495	0.487	0.486	0.484	0.451	0.465	0.450	0.461	0.457	0.477	0.519	0.524	0.521	0.500	0.471	0.493
	Avg.	**0.390**	**0.415**	0.418	0.431	0.439	0.442	0.412	0.428	0.428	0.433	0.407	0.421	0.439	0.448	0.427	0.441	0.415	0.431	0.404	0.420	0.419	0.437	0.440	0.463	0.488	0.452	0.425	0.439
ETTh2	96	**0.241**	**0.314**	0.270	0.331	0.283	0.342	0.279	0.343	0.285	0.342	0.263	0.329	0.297	0.348	0.281	0.351	0.283	0.345	0.263	0.332	0.274	0.337	0.611	0.557	0.334	0.370	0.289	0.353
	192	**0.282**	**0.345**	0.334	0.374	0.359	0.391	0.363	0.397	0.357	0.387	0.319	0.372	0.371	0.403	0.349	0.387	0.339	0.381	0.320	0.374	0.348	0.384	0.703	0.624	0.404	0.413	0.383	0.418
	336	**0.310**	**0.369**	0.376	0.405	0.361	0.401	0.381	0.419	0.369	0.402	0.311	0.373	0.404	0.428	0.366	0.413	0.366	0.405	0.313	0.376	0.377	0.416	0.827	0.675	0.389	0.435	0.448	0.465
	720	**0.392**	**0.429**	0.406	0.436	0.407	0.440	0.442	0.467	0.414	0.436	0.401	0.434	0.424	0.444	0.401	0.436	0.395	0.436	**0.392**	0.433	0.406	0.441	1.094	0.775	0.434	0.448	0.605	0.551
	Avg.	**0.306**	**0.364**	0.347	0.387	0.353	0.394	0.366	0.406	0.356	0.392	0.324	0.377	0.374	0.406	0.349	0.397	0.346	0.392	0.322	0.379	0.351	0.395	0.809	0.658	0.390	0.417	0.431	0.447
Weather	96	**0.140**	**0.180**	0.143	0.182	0.152	0.202	0.148	0.203	0.157	0.204	0.147	0.198	0.159	0.208	0.147	0.198	0.149	0.200	0.149	0.200	0.149	0.198	0.146	0.212	0.172	0.220	0.170	0.230
	192	**0.181**	**0.223**	0.185	0.224	0.205	0.251	0.193	0.243	0.205	0.248	0.194	0.238	0.200	0.248	0.192	0.243	0.193	0.240	0.196	0.245	0.194	0.241	0.195	0.261	0.219	0.261	0.216	0.273
	336	**0.231**	**0.263**	0.235	**0.263**	0.248	0.282	0.242	0.281	0.264	0.291	0.235	0.277	0.253	0.289	0.247	0.284	0.241	0.279	0.238	0.277	0.245	0.282	0.252	0.311	0.280	0.306	0.258	0.307
	720	**0.302**	**0.313**	0.308	0.319	0.341	0.351	0.315	0.332	0.346	0.344	0.308	0.331	0.321	0.338	0.318	0.330	0.324	0.338	0.314	0.334	0.314	0.334	0.318	0.363	0.365	0.359	0.323	0.362
	Avg.	**0.214**	**0.245**	0.218	0.247	0.237	0.271	0.225	0.265	0.243	0.271	0.221	0.261	0.233	0.271	0.226	0.264	0.227	0.264	0.224	0.264	0.226	0.264	0.228	0.287	0.259	0.287	0.242	0.293
ECL	96	**0.116**	**0.213**	0.128	0.218	0.127	0.222	0.131	0.228	0.143	0.237	0.142	0.223	0.138	0.237	0.142	0.247	0.134	0.228	0.129	0.226	0.129	0.222	0.135	0.237	0.168	0.272	0.140	0.237
	192	**0.136**	**0.231**	0.145	0.235	0.151	0.244	0.151	0.244	0.151	0.248	0.165	0.241	0.157	0.256	0.168	0.269	0.155	0.248	0.143	0.239	0.147	0.240	0.160	0.262	0.184	0.289	0.152	0.249
	336	**0.157**	**0.251**	0.162	**0.251**	0.167	0.260	0.167	0.262	0.176	0.269	0.164	0.269	0.167	0.264	0.171	0.260	0.173	0.268	0.161	0.259	0.163	0.259	0.182	0.282	0.198	0.300	0.170	0.267
	720	0.195	0.280	0.190	**0.278**	0.223	0.314	0.200	0.292	0.201	0.293	0.190	0.284	0.194	0.286	0.209	0.313	**0.186**	0.282	0.191	0.286	0.197	0.290	0.246	0.337	0.220	0.320	0.203	0.301
	Avg.	**0.151**	**0.244**	0.156	0.246	0.167	0.260	0.162	0.257	0.168	0.262	0.165	0.254	0.164	0.261	0.173	0.272	0.162	0.257	0.156	0.253	0.159	0.253	0.181	0.280	0.193	0.295	0.166	0.264
Solar	96	**0.166**	**0.195**	0.182	0.205	0.193	0.232	0.188	0.235	0.169	0.229	0.204	0.260	0.188	0.232	0.179	0.232	0.197	0.241	0.202	0.263	0.178	0.229	0.183	0.208	0.219	0.314	0.197	0.210
	192	**0.185**	**0.206**	0.198	0.217	0.207	0.261	0.203	0.246	0.189	0.252	0.227	0.272	0.193	0.248	0.201	0.259	0.231	0.264	0.223	0.279	0.189	0.246	0.208	0.226	0.231	0.322	0.218	0.222
	336	0.195	**0.213**	0.203	0.222	0.210	0.263	0.213	0.252	0.199	0.262	0.241	0.288	0.203	0.249	**0.190**	0.256	0.216	0.272	0.241	0.292	0.198	0.249	0.212	0.239	0.246	0.337	0.234	0.231
	720	0.204	**0.218**	**0.201**	0.223	0.223	0.278	0.215	0.257	0.207	0.259	0.242	0.287	0.223	0.261	0.203	0.261	0.250	0.281	0.247	0.292	0.209	0.256	0.215	0.256	0.280	0.363	0.243	0.241
	Avg.	**0.188**	**0.208**	0.196	0.217	0.208	0.258	0.205	0.248	0.191	0.251	0.229	0.277	0.202	0.248	0.193	0.252	0.224	0.265	0.228	0.282	0.194	0.245	0.205	0.232	0.244	0.334	0.223	0.226

Table B.1: Full results for the horizon-specific comparison. Results are reported as MSE and MAE for $H \in \{96, 192, 336, 720\}$, and Avg. is the arithmetic mean over the four horizons. ISCF-BSCA uses one unified model per dataset, whereas the baselines follow their horizon-specific evaluation protocols. The best and second-best displayed values are highlighted in bold and underlined, respectively.

Dataset	H	ISCF-BSCA (Ours)		TimeAlign (2026)		QDF (2026)		AMD (2025)		SimpleTM (2025)		iTransformer (2024b)		PatchTST (2023)		DLinear (2023)	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTm1	96	**0.265**	**0.326**	<u>0.290</u>	<u>0.338</u>	0.292	0.348	0.294	0.348	0.326	0.368	0.346	0.382	0.298	0.350	0.301	0.346
	192	**0.304**	**0.349**	<u>0.325</u>	<u>0.358</u>	0.330	0.371	0.333	0.368	0.362	0.385	0.384	0.399	0.335	0.372	0.336	0.367
	336	**0.343**	**0.371**	<u>0.355</u>	<u>0.378</u>	0.364	0.391	0.366	0.387	0.392	0.406	0.418	0.420	0.367	0.392	0.371	0.387
	720	**0.404**	**0.405**	<u>0.408</u>	<u>0.409</u>	0.423	0.424	0.423	0.417	0.454	0.438	0.487	0.456	0.415	0.421	0.427	0.423
	Avg.	**0.329**	**0.363**	<u>0.345</u>	<u>0.371</u>	0.352	0.384	0.354	0.380	0.384	0.399	0.409	0.415	0.354	0.384	0.359	0.381
ETTm2	96	**0.159**	**0.242**	0.168	<u>0.253</u>	<u>0.165</u>	0.255	0.170	0.260	0.186	0.270	0.190	0.277	0.169	0.258	0.189	0.289
	192	<u>0.218</u>	**0.285**	**0.216**	<u>0.286</u>	0.223	0.294	0.222	0.295	0.247	0.308	0.253	0.315	0.223	0.295	0.256	0.339
	336	<u>0.270</u>	**0.318**	**0.268**	<u>0.320</u>	0.276	0.330	0.271	0.327	0.305	0.344	0.318	0.354	0.275	0.329	0.320	0.381
	720	**0.343**	**0.371**	<u>0.346</u>	<u>0.374</u>	0.364	0.385	0.355	0.381	0.401	0.399	0.416	0.409	0.365	0.383	0.403	0.428
	Avg.	**0.248**	**0.304**	<u>0.250</u>	<u>0.308</u>	0.257	0.316	0.255	0.316	0.285	0.330	0.294	0.339	0.258	0.316	0.292	0.359
ETTh1	96	**0.348**	**0.386**	<u>0.375</u>	0.404	0.401	0.418	0.378	<u>0.403</u>	0.382	0.406	0.409	0.422	0.394	0.414	0.416	0.436
	192	**0.377**	**0.403**	0.411	0.425	0.441	0.441	<u>0.410</u>	<u>0.424</u>	0.434	0.431	0.457	0.448	0.429	0.433	0.457	0.465
	336	**0.393**	**0.414**	0.433	0.437	0.467	0.459	<u>0.426</u>	<u>0.433</u>	0.448	0.442	0.496	0.467	0.440	0.442	0.484	0.481
	720	**0.443**	**0.458**	0.459	0.464	0.499	0.493	<u>0.452</u>	<u>0.461</u>	0.470	0.465	0.509	0.494	0.455	0.469	0.538	0.538
	Avg.	**0.390**	**0.415**	0.420	0.433	0.452	0.453	<u>0.416</u>	<u>0.430</u>	0.434	0.436	0.468	0.458	0.429	0.440	0.474	0.480
ETTh2	96	**0.241**	**0.314**	0.296	0.349	<u>0.272</u>	0.340	0.285	0.351	0.285	0.344	0.303	0.353	0.277	<u>0.339</u>	0.322	0.382
	192	**0.282**	**0.345**	0.364	0.393	<u>0.334</u>	<u>0.382</u>	0.374	0.405	0.358	0.388	0.385	0.401	0.345	<u>0.382</u>	0.367	0.408
	336	**0.310**	**0.369**	0.405	0.418	<u>0.358</u>	<u>0.404</u>	0.379	0.421	0.367	<u>0.404</u>	0.424	0.433	0.379	0.414	0.385	0.425
	720	**0.392**	**0.429**	<u>0.406</u>	<u>0.436</u>	0.407	0.440	0.442	0.467	0.414	<u>0.436</u>	0.430	0.446	0.407	0.443	0.584	0.543
	Avg.	**0.306**	**0.364**	0.368	0.399	<u>0.343</u>	<u>0.392</u>	0.370	0.411	0.356	0.393	0.385	0.408	0.352	0.395	0.415	0.439
Weather	96	**0.140**	**0.180**	<u>0.148</u>	<u>0.187</u>	0.191	0.246	0.149	0.204	0.177	0.222	0.179	0.222	0.155	0.205	0.177	0.240
	192	**0.181**	**0.223**	<u>0.187</u>	<u>0.226</u>	0.231	0.276	0.194	0.244	0.221	0.259	0.228	0.262	0.199	0.246	0.218	0.278
	336	**0.231**	**0.263**	<u>0.238</u>	<u>0.268</u>	0.276	0.307	0.244	0.282	0.273	0.296	0.283	0.301	0.249	0.284	0.264	0.317
	720	**0.302**	**0.313**	<u>0.308</u>	<u>0.319</u>	0.341	0.351	0.315	0.332	0.346	0.344	0.359	0.351	0.319	0.335	0.328	0.369
	Avg.	**0.214**	**0.245**	<u>0.220</u>	<u>0.250</u>	0.260	0.295	0.226	0.266	0.254	0.280	0.262	0.284	0.231	0.268	0.247	0.301
ECL	96	**0.116**	**0.213**	<u>0.129</u>	<u>0.220</u>	0.155	0.258	0.132	0.230	0.150	0.246	0.152	0.245	0.132	0.228	0.140	0.238
	192	**0.136**	**0.231**	<u>0.145</u>	<u>0.235</u>	0.169	0.269	0.152	0.246	0.152	0.251	0.164	0.257	0.148	0.243	0.154	0.251
	336	**0.157**	**0.251**	<u>0.163</u>	<u>0.252</u>	0.185	0.284	0.166	0.261	0.173	0.269	0.179	0.273	0.164	0.259	0.169	0.268
	720	<u>0.195</u>	<u>0.280</u>	**0.190**	**0.278**	0.223	0.314	0.200	0.292	0.201	0.293	0.211	0.301	0.201	0.291	0.204	0.301
	Avg.	**0.151**	**0.244**	<u>0.157</u>	<u>0.246</u>	0.183	0.281	0.163	0.257	0.169	0.265	0.177	0.269	0.161	0.255	0.167	0.265
Solar	96	**0.166**	**0.195**	0.179	<u>0.211</u>	0.186	0.256	0.190	0.242	<u>0.171</u>	0.233	0.197	0.239	0.181	0.235	0.222	0.292
	192	**0.185**	**0.206**	0.192	<u>0.218</u>	0.205	0.267	0.202	0.249	<u>0.187</u>	0.246	0.228	0.259	0.190	0.245	0.250	0.309
	336	<u>0.195</u>	**0.213**	0.198	<u>0.223</u>	0.215	0.272	0.210	0.254	**0.193**	0.251	0.243	0.271	0.198	0.251	0.269	0.324
	720	<u>0.204</u>	**0.218**	**0.201**	<u>0.223</u>	0.223	0.278	0.215	0.257	0.207	0.259	0.250	0.276	0.206	0.257	0.271	0.327
	Avg.	**0.188**	**0.208**	0.193	<u>0.219</u>	0.207	0.268	0.204	0.250	<u>0.190</u>	0.247	0.229	0.261	0.194	0.247	0.253	0.313

Table B.2: Full results for the one-model-all-horizons comparison. Each method uses one maximum-horizon model per dataset, and shorter horizons are evaluated from the corresponding output prefixes. Results are reported as MSE and MAE, and Avg. denotes the arithmetic mean over $H \in \{96, 192, 336, 720\}$. The best and second-best displayed values are highlighted in bold and underlined, respectively.

Appendix C. Visualization

Fig. C.1 presents supplementary varied-horizon forecasting examples for the seven paper-core datasets. For each dataset, two validation samples show the ground-truth future and the fused ISCF-BSCA prediction over the maximum target length $T = 720$. The paired rulers and vertical guides mark the four supported prediction lengths, showing how the same predicted trajectory provides the prefixes requested at $H \in \{96, 192, 336, 720\}$.

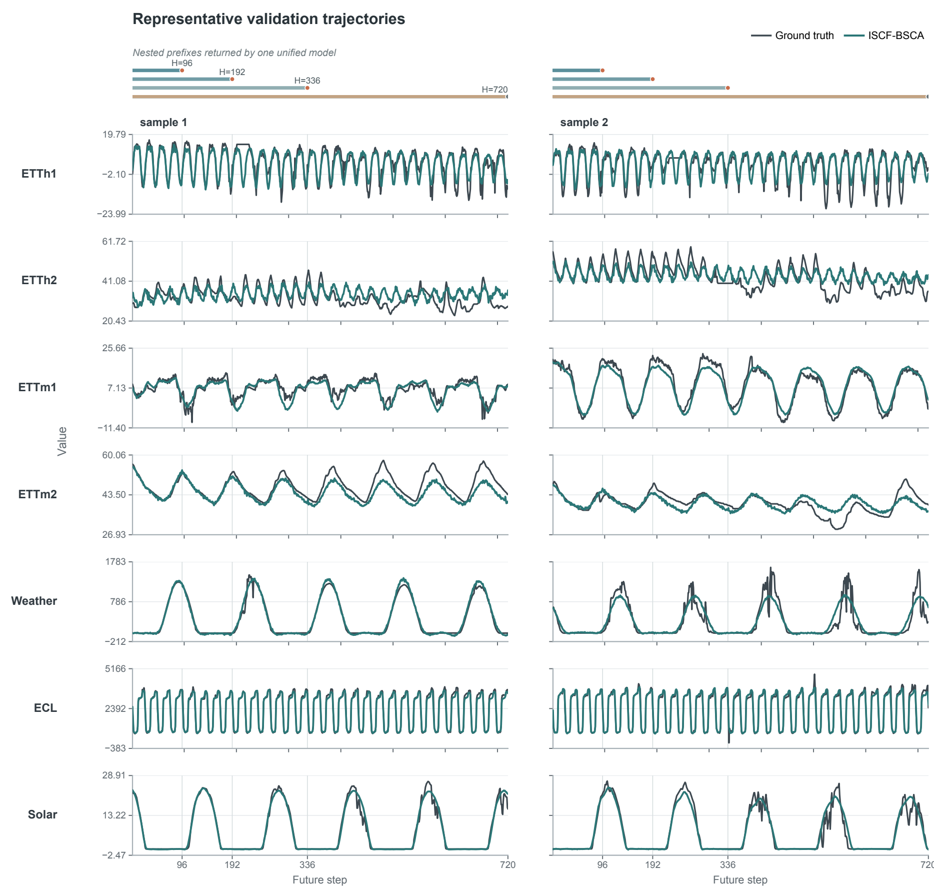

Figure C.1: Visualization of varied-horizon forecasts across seven datasets. The dark curve denotes the ground truth and the teal curve denotes the prediction from the frozen unified ISCF-BSCA model. Two validation examples are shown for each dataset. The paired rulers and vertical guides identify the four supported prediction lengths.